AUTOSAR High-Level Design (HLD) Document

Subsystem: Braking, ADAS & Automated Lateral Control | Version: 2.0

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1.0 System Architecture Overview & Scope (Revision 2.0)

This revised High-Level Design (HLD v2.0) incorporates Autonomous Driving Assistance (ADAS Level 2+) features, upgrading the braking and longitudinal dynamics pipeline to support Adaptive Cruise Control (ACC) and Automated Emergency Steering (AES). Legacy CAN-FD gateway architecture is replaced with a high-throughput Automotive Ethernet Backbone.

1.1 Updated Software Components Definition

• **BrakeController**: Application SWC (ASIL-D). Enhanced with regenerative brake blending and autonomous emergency braking (AEB) trigger.
• **VehicleController**: Application SWC (ASIL-D). Central motion supervisor with multi-actuator torque vectoring.
• **SensorFusion**: Application SWC (ASIL-D - *Upgraded from ASIL-B*). Tri-modal perception merging radar, vision, and ultrasonic tracks.
• **RadarInterface**: Sensor SWC (ASIL-B). Upgraded for 77GHz imaging radar.
• **CameraInterface**: [NEW] Sensor SWC (ASIL-B). Vision object classification and lane marker detection.
• **ADASCruiseControl**: [NEW] Application SWC (ASIL-C). Calculates velocity profiles, gap regulation, and headway time.
• **SafetyController**: Complex SWC (ASIL-D). Enhanced multi-core safe lockstep monitor.
• *(Note: Legacy CAN_Gateway component removed in v2.0 in favor of EthernetSwitchSWC).*

2.0 Revised Software Component Port Specifications

Component (SWC)	Type	ASIL	Provided Ports (P-Port)	Required Ports (R-Port)
BrakeController	Application SWC	ASIL-D	P_BrakeCommand, P_BrakeStatus, P_AEB_State	R_VehState, R_ADAS_DecelReq
VehicleController	Application SWC	ASIL-D	P_VehState, P_TargetTorque	R_DriverDemand, R_BrakeStatus, R_ADAS_Moti
SensorFusion	Application SWC	ASIL-D	P_FusionTrack, P_ObstacleDist, P_LaneGeometry	R_RadarRaw, R_CameraVisionRaw
RadarInterface	Sensor SWC	ASIL-B	P_RadarRaw	R_EthData_Radar
CameraInterface	Sensor SWC	ASIL-B	P_CameraVisionRaw	R_EthData_Camera
ADASCruiseControl	Application SWC	ASIL-C	P_ADAS_DecelReq, P_ADAS_MotionReq	R_FusionTrack, R_VehState
SafetyController	Complex SWC	ASIL-D	P_SafeStateCmd	R_BrakeCommand, R_VehState, R_ADAS_Dece

3.0 Updated Interfaces & Signal Routing

• **ADAS_DecelRequestInterface**: [NEW] Source **ADASCruiseControl** -> Target **BrakeController**. Signals: TargetDecelerationReq (float32), UrgencyLevel (uint8), AEB_Arm (boolean).
• **BrakeCommandInterface**: Source **BrakeController** -> Target **VehicleController**. Signals: TargetDeceleration (float32), BrakePedalStatus (uint8), EBD_Active (boolean), RegBrakeTorque (float32).
• **CameraVisionStreamInterface**: [NEW] Source **CameraInterface** -> Target **SensorFusion**. Signals: ObjectBoundingBoxes, LaneCurvature, PedestrianDetectionFlag.
• **SensorTelemetryInterface**: Source **SensorFusion** -> Target **ADASCruiseControl**. Signals: LeadVehicleDist, LeadVehicleSpeed, TargetCutInProb.
• **SafetySupervisionInterface**: Source **SafetyController** -> Target **BrakeController**. Signals: SafeStateActive (boolean), OverrideBrakeLevel (uint8).

4.0 Potential Inconsistency & Safety Notes (For Review)

Architecture Observation: On Page 2, ADASCruiseControl is designated ASIL-C while its output connects directly to BrakeController (ASIL-D) via ADAS_DecelRequestInterface. Furthermore, in Section 3.0, ADASCruiseControl interfaces with BrakeController, but legacy reference on Page 3 still suggests direct override to SafetyController. This requires formal Human Engineer Review and Safety Manager signoff.

5.0 Execution Schedule & Ethernet Latency Budgets

- **CameraInterface_Step (30ms Periodic)**: Processes vision frame inference metadata.
- **ADASCruiseControl_Step (10ms Periodic)**: Headway distance control loop.
- **BrakeController_Step (10ms Periodic)**: Closed-loop deceleration actuation.